Alexander Paschal

CONTACT INFORMATION
E-mail: alexander.paschal@ens-lyon.fr
Website: alexpasch.al

EDUCATION

École Normale Supérieure de Lyon, Lyon, France

M2, Mathematics, August 2025–September 2026 (expected). SFRI MathInfi scholarship recipient. Thesis advisor: Filippo Santambrogio.

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

B.S., Mathematics, August 2023–May 2025. Completed coursework.

RESEARCH EXPERIENCE

Université Claude Bernard Lyon 1, Lyon, France

Optimal transport. March 2026–Present. M2 internship. Mentored by Professor Filippo Santambrogio.

The Ohio State University, Columbus, Ohio

Ergodic theory. June 2024–Present. ROMUS and tOSU-IM program participant. Mentored by Professor Daniel Thompson.

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Analytic number theory. October 2023–May 2025. Mentored by Professor Idris Assani.

PUBLICATIONS

[1] Alexander Paschal. Local Regularity Properties for Potentials on Topological Markov Shifts. *In preparation*.

[2] Alexander Paschal, Amy Somers, and Daniel Thompson. The Variational Principle for Pressure of Countable State Shift Spaces With Weak Specification. *In preparation*.

[3] Alexander Paschal and Amy Somers. The Variational Principle for Entropy of Countable State Shift Spaces With Specification. *Accepted*, *PUMP Journal of Undergraduate Research*, available here.

[4] Idris Assani, Aiden Chester, and Alexander Paschal. On Robin's Inequality and the Kaneko-Lagarias Inequality. *In revision*, *Integers*, available here.

CONFERENCE TALKS

[5] 58th Spring Topology and Dynamics Conference, Christopher Newport University, March 6–8, 2025, Newport News, Virginia.

SEMINAR TALKS

[6] EYAWKAJKOS Seminar, Université Claude Bernard Lyon 1, June 17, 2026, Lyon, France.

[7] Ergodic Theory Seminar, The Ohio State University, July 23, 2025, Columbus, Ohio.

CONFERENCE POSTERS

[8] International Conference on Dynamical Systems, Instituto de Matemática Pura e Aplicada, August 24–28, 2026, Rio de Janeiro, Brazil.

[9] New Directions in Thermodynamic Formalism, Centre International de Rencontres Mathématiques, June 22–26, 2026, Marseille, France.

[10] Midwest Dynamical Systems Conference, University of Notre Dame, May 2–3, 2026, Notre Dame, Indiana. Poster available here.

[11] ISTA Summer School in Dynamical Systems, Bundesinstitut für Erwachsenenbildung, September 22—26, 2025, Strobl, Austria.

[12] Beyond Uniform Hyperbolicity Conference, International Centre for Theoretical Physics, June 2–13, 2025, Trieste, Italy.

[13] Midwest Dynamical Systems Conference, Northwestern University, April 25–27, 2025, Evanston, Illinois.

CONFERENCES ATTENDED

[14] Recent Trends in Ergodic Theory, Symbolic Dynamics, and Connections to Combinatorics, Northwestern University, August 31–September 4, 2026, Evanston, Illinois.

[15] 40th Summer Conference on Topology and its Applications, July 13–17, 2025, Split, Croatia.

[16] Journées IA, Preuve, et Formalisation, École Normale Supérieure de Lyon, June 16–17, 2026, Lyon, France.

[17] Everything You Always Wanted to Know About Keller-Segel and Aggregation-Diffusion, Institut Camille Jordan, April 23–24, 2026, Lyon, France.

[18] Première Rencontre Nationale du RT Optimisation, Institut Camille Jordan, November 26–28, 2025, Lyon, France.

[19] Young Mathematicians Conference, The Ohio State University, July 30–August 1, 2025, Columbus, Ohio. *Gave a short talk.*

[20] Joint Mathematics Meeting, January 8–11, 2025, Seattle, Washington. *Gave a short talk.*

[21] Regional Mathematics and Statistics Conference, University of North Carolina Greensboro, November 8–9, 2024, Greensboro, North Carolina. *Gave a short talk.*

[22] Young Mathematicians Conference, The Ohio State University, August 13–15, 2024, Columbus, Ohio. *Gave a short talk.*

EXPOSITORY WRITING

[23] Alexander Paschal. Thermodynamic Formalism in Holomorphic Dynamics. *Notes for a minicourse*, available here.

[24] Alexander Paschal. Carolina Math Club Logic Talks Notes. *Lecture notes*, available here.

EXPOSITORY TALKS

[25] Alexander Paschal. "Thermodynamic Formalism for Local Bowen Potentials on Countable-State Shift Spaces." New Directions in Thermodynamic Formalism, Centre International de Rencontres Mathématiques, June 22–26, 2026, Marseille, France. *Lightning talk*.

[26] Alexander Paschal. "Thermodynamic Formalism in Countable State Symbolic Dynamics." tOSU-IM Symposium, The Ohio State University, June 27, 2025, Columbus, Ohio.

[27] Alexander Paschal and Amy Somers. "The Variational Principle for Pressure of Countable State Shift Spaces With Specification." Beyond Uniformy Hyperbolicity Conference, International Centre for Theoretical Physics, June 2–13, 2025, Trieste, Italy. *Lightning talk*.

[28] Alexander Paschal. "Gödel's First Incompleteness Theorem." Carolina Math Club, March 4, 2025, Chapel Hill, NC. Notes available here.

[29] Alexander Paschal. "Paradox in Logical Systems." Carolina Math Club, February 28, 2025, Chapel Hill, NC. Notes available here.

[30] Alexander Paschal. "What is Symbolic Dynamics?" Carolina Math Club, August 17, 2024, Chapel Hill, NC.

[31] Alexander Paschal and Amy Somers. "The Variational Principle for Entropy of Countable State Shift Spaces With Specification." Consortium of Summer Undergraduate Research Experiences, The Ohio State University, July 25, 2024, Columbus, Ohio.

[32] Alexander Paschal and Amy Somers. "Entropy in Countable State Symbolic Dynamics." ROMUS Symposium, The Ohio State University, July 3, 2024, Columbus, Ohio.

[33] Alexander Paschal. "Superabundant Numbers and the Riemann Hypothesis." Carolina Math Club, April 1, 2024, Chapel Hill, NC.

SERVICE

University of North Carolina at Chapel Hill, Chapel Hill, NC

Undergraduate Learning Assistant

- MATH 522 (Advanced Calculus II) (Spring 2025)
- MATH 521 (Advanced Calculus I) (Spring 2024 and Fall 2024)
- MATH 381 (Discrete Mathematics) (Fall 2023)

Carolina Math Club

- President (Spring 2025)
- Academic Chair (Spring 2024 and Fall 2024)

Academic program review student representative (Fall 2024)

TECHNOLOGY

Proficient: Python, LaTeX, Wolfram Language (Mathematica).
Familiar: Java, Rust, Javascript.